

[DC] Beyond Visual Dominance: Haptics as Social Cues for Enhanced Multi-User Experiences in VR

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ABSTRACT

This research investigates the potential of haptic patterns serving as a social cue that supports users' social cognition in visually demanding virtual reality (VR) environments. While prior work in social VR has primarily focused on visual and auditory information, this research aims to examine how vibrotactile patterns can represent relational or group-relevant information and influence users' social understanding. Building on multisensory social cognition theories, the research aims to establish design principles for haptic-based social signals and explore how they complement visual cues in social VR contexts.

Index Terms: Haptic Pattern, Social Cognition, Virtual Reality.

1 INTRODUCTION AND BACKGROUND

Humans interact and comprehend social contexts through a variety of sensory modalities, including vision, audition, touch, and olfaction [1]. Virtual Reality (VR) has garnered significant attention since it transcends traditional two-dimensional video-based communication by presenting richer social information within three-dimensional environments. Prior VR research has explored various approaches to convey users' intentions and actions to remote collaborators, such as leveraging gaze-tracking information, manipulating vocal and visual characteristics, and employing cues like sketching or pointing gestures [9].

However, due to the inherently high visual dependency of VR [11], the majority of studies have focused primarily on visual or auditory cues. Although visual stimuli remain effective given their intuitive information delivery and sensory dominance [6], over-reliance on vision without considering other sensory channels may produce counterproductive effects. For instance, in fast-paced collaborative environments involving short-cycle repetitive tasks, densely populated virtual-crowd scenarios, or multiplayer gaming where visual effects overlap, effectively leveraging visual cues becomes challenging. Consequently, it is necessary to develop alternative sensory-based social cues that can complement visual information.

Haptic cues have been widely adopted as a supplementary modality to compensate for visual limitations or to assist people with visual impairments, serving as guidance cues in contexts such as Braille reading and navigation [12]. In VR environments, haptics have also been utilized to simulate the physical contact with remote users [10]. Although recent multisensory research has begun incorporating haptic feedback alongside audiovisual stimuli in VR, this body of work has predominantly focused on enhancing task performance in collaborative settings rather than examining the socio-cognitive effects of such cues [7].

It is particularly important to investigate whether haptic stimuli can function as social cues capable of conveying relational information, such as interpersonal dynamics, intimacy, and a sense of

belonging, beyond merely replicating the sensation of touch or presenting physical information. Furthermore, it is essential to examine how haptic cues influence users' social cognition when used as social signals. Therefore, I define the following research questions:

- RQ1** Can haptic patterns be utilized to modulate users' social impression of others?
- RQ2** Can relationship information be conveyed not only at the individual level but also at the group level?
- RQ3** What haptic pattern designs most effectively convey social information?

2 RESEARCH PROGRESS

To address **RQ1**, I first drew upon the well-established finding that people are attracted to those who are similar to themselves [8]. Just as users express themselves through avatars in social VR, I sought to examine whether assigning users a haptic identity, a distinctive pattern that represents the self, could influence their perception of others based on the relationship between patterns. Prior to conducting this study, it was crucial to select and design haptic stimuli with high perceptual resolution capable of conveying nuanced social information. Building on prior research, I focused on vibrotactile rhythm, composed of combinations of frequency and duration, and developed a pattern dataset based on four-beat rhythmic structures. I then quantified the perceived similarity among selected patterns using pairwise dissimilarity ratings and multidimensional scaling (MDS).

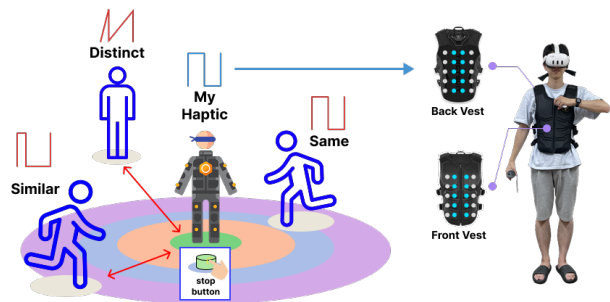


Figure 1: The experimental overview of the first study. Participants experienced their own haptic and the assigned haptic pattern of the approaching agents via their haptic vest. Depending on participants' subjective comfort, they could press the button to stop the agents' movement.

In a user study (N=33), participants first selected a pattern to represent themselves and underwent an embodiment phase to internalize it. They then evaluated their impressions of virtual agents, each assigned a different haptic pattern. As an agent approached the participant, they experienced the agent's pattern through a wearable haptic suit. Participants were asked to stop the agent's movement at the distance at which they felt most comfortable (Figure 1).

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The results revealed that when the haptic patterns assigned to participants and agents were more similar, participants perceived the agents as more intimate and felt a greater sense of connection, while also reporting shorter interpersonal distances [5]. This study served as a milestone, demonstrating that the relationships between haptic patterns can function as social markers to influence users' social perception.

3 RESEARCH PLAN

Having established in previous research that haptic patterns can influence impressions of individual others, I now aim to extend this scope by examining the effects of haptic patterns at the group level. Addressing **RQ2**, this extension is particularly important in social contexts because people define themselves according to the groups to which they belong, and again, they project their self-perception onto those groups [2]. In addition, it is necessary to investigate how haptic patterns interact with the presence or absence of visual cues. Since sensory information is typically presented as multisensory rather than unisensory input, examining the effects of multisensory stimuli compared to unisensory stimuli is essential.

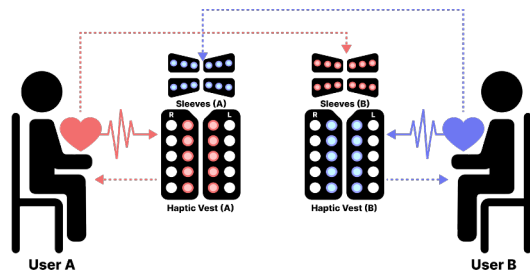


Figure 2: Brief overview of the subsequent research plan. Each participant's heart rate signal is translated into haptic feedback and delivered through their haptic suit. Simultaneously, their heart rate signals are conveyed to their partner via haptic sleeves, enabling the sharing of physiological states.

Furthermore, while the previous study employed four-beat on-beat vibrotactile rhythms based on prior research, exploring more complex pattern structures or examining the effects of haptic feedback grounded in physiological signals would represent an important direction for addressing **RQ3**. In this regard, interoception refers to the perception of bodily signals originating from inside the body. Existing research suggests that interoceptive ability is associated with social presence in interpersonal contexts [3, 4]. If bodily signals from others or from one's own body could be exteroceptively translated through haptic stimuli, it would be worthwhile to examine whether such physiologically related cues can serve as effective social signals in VR. Moreover, comparing whether these physiologically grounded patterns function more effectively than simple rhythm-based patterns would provide more comprehensive insights into designing haptic-based social cues in VR environments (Figure 2).

4 CONCLUSION AND PERSPECTIVES

This research explores the potential of using the haptic modality as a social cue to intuitively convey social information and examines its effects on VR users' social cognition. Rather than merely employing haptic patterns as cues that supplement or replace visual information, this research seeks to investigate social interaction methods that can be uniquely afforded through the haptic modality. Furthermore, through the lenses of social presence, social connectedness, intergroup perception, and interoception-exteroception, this work examines how social information conveyed through haptic cues influences users' social cognition across individual and group

contexts, as well as dynamic interaction scenarios in VR. To further expand and strengthen the contributions of this research, it is essential to explore and validate diverse haptic cue designs. Considering future research directions, I seek to engage in more in-depth discussions regarding haptic design grounded in users' physiological signals. Therefore, as part of the Doctoral Consortium, I anticipate discussing the following topics:

1. How can physiological signals be effectively translated into haptic feedback while controlling for noise? Are there methods or scenarios that enable robust translation even in dynamic situations?
2. How should haptic cues be designed to convey physiological information, and what design rationale should guide this process?

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